

Modeling, State-Feedback Control, and Observer Design for a Two-Wheeled Self-Balancing Robot

Vahid Danesh

Department of Mechanical Engineering

Stony Brook University

MEC 560 — Advanced Control Systems, Spring 2026

Instructor: Amin Fakhari, Ph.D.

Abstract—This report presents a complete control design study for a two-wheeled self-balancing robot (TWSBR) modeled as an underactuated inverted pendulum on a wheeled cart. The nonlinear equations of motion are derived from first principles using the Lagrangian formulation and linearized about the unstable upright equilibrium. Open-loop analysis confirms that the system is controllable, observable, and has a minimal realization of order four. A state-feedback controller is designed using the Linear Quadratic Regulator (LQR) method for stabilization and reference tracking. Full-order and reduced-order Luenberger observers are designed to estimate the unmeasured velocity states. Observer-based closed-loop stability is verified through the Separation Principle. Robustness under parameter uncertainty and actuator saturation is evaluated through nonlinear simulations in Python. All source code is publicly available here.

I. INTRODUCTION

Two-wheeled self-balancing robots (TWSBR) are dynamically unstable by nature. They require active feedback control to maintain the upright position. Their mechanical simplicity and rich control challenges make them a standard benchmark in control engineering research [1], [2].

The system belongs to the class of underactuated mechanical systems. It has two degrees of freedom but only one independent control input. This makes stabilization nontrivial and motivates the use of modern state-space control methods.

This work applies linear systems theory to the full control design pipeline including modeling, stability analysis, controller design, observer design, and simulation-based evaluation. All methods follow the framework of Hespanha [3].

A. Related Work

Grasser et al. introduced JOE, one of the earliest self-balancing two-wheeled robots [1]. They derived the equations of motion using Newton-Euler mechanics and designed a state-feedback controller in continuous time.

Kim and Kwon identified systematic modeling errors in prior works and provided a rigorous derivation using both the Lagrangian approach and Kane's method [2].

Frankovsky et al. extended the standard mechanical model to include DC gearmotor dynamics, providing a more complete actuator model [4].

Shishaye demonstrated a full control design pipeline for the inverted pendulum system, covering pole placement, LQR, tracking control, and observer design in a student-oriented framework [5].

Choudhry et al. addressed robustness concerns for the TWSBR under parameter uncertainty and external disturbances [6].

Petcu and Andreescu developed LQR control with a Luenberger state observer for vertical-angle stabilization and position tracking, alongside a PID controller for angle-only stabilization [7].

Finally, a number of student projects and theses have built and controlled physical TWIP platforms [8]–[11].

B. Project Scope

This report follows the three-phase structure of the MEC 560 final project. Phase I derives the nonlinear equations of motion from first principles and linearizes the system about the upright equilibrium. The open-loop model is then analyzed for stability, controllability, observability, and minimal realization.

Phase II covers controller and observer design. An LQR state-feedback controller is designed for stabilization and extended with integral action for reference tracking. Full-order and reduced-order Luenberger observers are designed to estimate the unmeasured velocity states. Combined observer-based stability is verified using the Separation Principle.

Phase III evaluates the complete system through nonlinear simulations in Python. Performance is assessed across multiple initial conditions. Robustness under parameter uncertainty and the effect of actuator saturation are also examined.

II. PRELIMINARIES

A. State-Space Representation

A linear time-invariant (LTI) system is written as:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u, \quad (1)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}u, \quad (2)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the state vector, $u \in \mathbb{R}^m$ is the input, $\mathbf{y} \in \mathbb{R}^p$ is the output, and $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$ are constant matrices of appropriate dimensions [3].

B. Stability

An equilibrium $\mathbf{x}^*$ of the nonlinear system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u)$ is Lyapunov stable if trajectories starting near $\mathbf{x}^*$ remain near it. It is asymptotically stable if they also converge to $\mathbf{x}^*$. For the linearized system (1), asymptotic stability requires all eigenvalues of $\mathbf{A}$ to lie in the open left-half plane (LHP) [3].

C. Controllability and Stabilizability

The system (A, B) is controllable if the controllability matrix

$$\mathcal{C} = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B] \quad (3)$$

has full row rank [3]. The system is stabilizable if all unstable modes are controllable, which is a weaker condition sufficient for closed-loop stabilization.

D. Observability and Detectability

The system (A, C) is observable if the observability matrix

$$\mathcal{O} = [C^\top \quad A^\top C^\top \quad \dots \quad (A^\top)^{n-1} C^\top]^\top \quad (4)$$

has full column rank [3]. The system is detectable if all unstable modes are observable.

E. Minimal Realization

A state-space realization (A, B, C, D) is minimal if and only if the system is both controllable and observable [3]. The order of a minimal realization equals the number of poles of the transfer function matrix.

F. Linear Quadratic Regulator

The LQR method finds the state-feedback gain K that minimizes the infinite-horizon cost:

$$J = \int_0^\infty (x^\top Q x + u^\top R u) dt, \quad (5)$$

where $Q \succeq 0$ penalizes state deviations and $R \succ 0$ penalizes control effort [3]. The optimal control law is $u = -Kx$, where $K = R^{-1}B^\top P$ and $P \succ 0$ solves the algebraic Riccati equation:

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0. \quad (6)$$

G. Luenberger Observer

When not all states are measurable, a full-order observer estimates the full state vector:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}), \quad (7)$$

where L is the observer gain matrix [3]. The estimation error $e = x - \hat{x}$ evolves as $\dot{e} = (A - LC)e$. The observer is stable if all eigenvalues of $(A - LC)$ are in the LHP.

H. Separation Principle

When combining a state-feedback controller $u = -K\hat{x}$ with an observer (7), the closed-loop poles are the union of the eigenvalues of $(A - BK)$ and $(A - LC)$ [3]. The gains K and L can therefore be designed independently.

III. PROBLEM STATEMENT

A. System Description

The two-wheeled self-balancing robot is modeled as a rigid body (the pendulum) mounted on a wheeled cart, moving along a single horizontal axis. Fig. 1 shows the system schematic. The wheels are driven by stepper motors, modeled as a single equivalent horizontal force u applied at the wheel-ground contact point. The robot has two degrees of freedom: the cart position x and the body tilt angle θ . Since only one input is available, the system is underactuated.

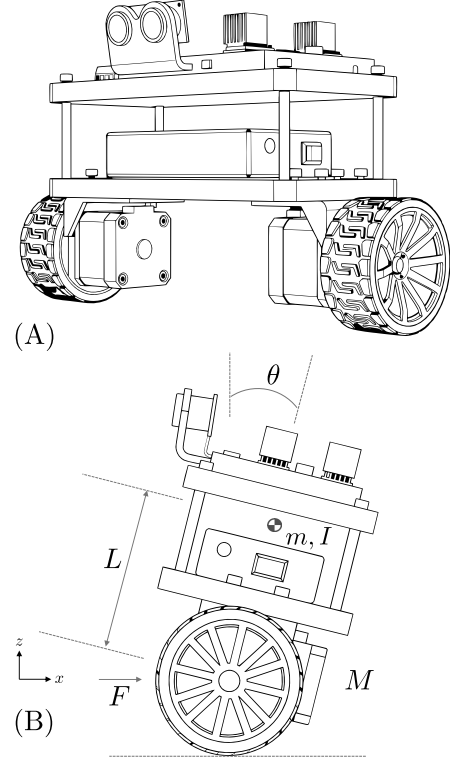


Fig. 1. (A) Schematic of the two-wheeled self-balancing robot. (B) Modeled as an inverted pendulum on a wheeled cart.

B. State Variables and Parameters

The state vector is defined in the $[\mathbf{q}; \dot{\mathbf{q}}]$ block convention, with generalized coordinates $\mathbf{q} = [x, \theta]^\top$:

$$\mathbf{x} = [x \quad \theta \quad \dot{x} \quad \dot{\theta}]^\top, \quad (8)$$

where x is the wheel base position (m), θ is the tilt angle from vertical (rad), $\dot{x}$ is the translational velocity (m/s), and $\dot{\theta}$ is the angular rate (rad/s). This ordering places the measured outputs (x and θ) as the first two states, giving $\mathbf{C} = [\mathbf{I}_2 \mid \mathbf{0}_2]$.

The system output is:

$$\mathbf{y} = [x \quad \theta]^\top, \quad (9)$$

corresponding to measurements from a wheel encoder and an inertial measurement unit (IMU), respectively.

The physical parameters are listed in Table I.

TABLE I
SYSTEM PARAMETERS

Symbol	Description	Value	Unit
M	Wheel and base mass	0.5	kg
m	Body (pendulum) mass	0.2	kg
L	Wheel axis to body CoM	0.1	m
I	Body moment of inertia	0.006	kg·m ²
b	Rolling friction	0.1	N·m·s
g	Gravitational acceleration	9.81	m/s ²

C. Control Objectives

The control objectives are:

- 1) **Stabilization:** Maintain $\theta \approx 0$ from any small initial perturbation.
- 2) **Position tracking:** Drive x to a desired reference $x_{\text{ref}}(t)$ with zero steady-state error.
- 3) **Observer convergence:** Estimate all states accurately from $\mathbf{y}$ alone, with estimation error converging to zero.
- 4) **Robustness:** Maintain stability under $\pm 20\%$ parameter uncertainty and actuator saturation.

D. Performance Specifications

The target closed-loop performance is: rise time for x less than 1 s, settling time less than 5 s, overshoot in θ less than 20° (0.35 rad), and zero steady-state error.

IV. METHODOLOGY

A. Lagrangian Derivation

The total kinetic energy of the system is:

$$T = \frac{1}{2}(M+m)\dot{x}^2 + mL\dot{x}\dot{\theta}\cos\theta + \frac{1}{2}(I+mL^2)\dot{\theta}^2. \quad (10)$$

The potential energy is:

$$V = mgL\cos\theta. \quad (11)$$

Applying the Euler-Lagrange equations with generalized coordinates (x, θ) and generalized forces $(u, 0)$ yields two coupled nonlinear ordinary differential equations.

For the cart coordinate x :

$$(M+m)\ddot{x} + mL\ddot{\theta}\cos\theta - mL\dot{\theta}^2\sin\theta + b\dot{x} = u. \quad (12)$$

For the tilt coordinate θ :

$$(I+mL^2)\ddot{\theta} + mL\ddot{x}\cos\theta - mgL\sin\theta = 0. \quad (13)$$

These two equations define the nonlinear state-space model $\dot{\mathbf{x}} = f(\mathbf{x}, u)$.

B. Equilibrium Analysis

Setting $f(\mathbf{x}^*, u^*) = \mathbf{0}$ yields the equilibrium conditions. All velocity states must be zero: $\dot{x}^* = \dot{\theta}^* = 0$. From (13), the tilt must satisfy $\sin\theta^* = 0$, giving $\theta^* \in \{0, \pi\}$.

The upright position $\theta^* = 0$ is the operating equilibrium. The hanging position $\theta^* = \pi$ is stable but physically irrelevant. For both equilibria, the required input is $u^* = 0$. The cart position x^* is a free parameter due to translation invariance; we set $x^* = 0$ without loss of generality.

C. Jacobian Linearization

The system is linearized about $(\mathbf{x}^*, u^*) = (\mathbf{0}, 0)$ using a first-order Taylor expansion. Applying the small-angle approximations $\sin\theta \approx \theta$, $\cos\theta \approx 1$, and neglecting the $\dot{\theta}^2 \sin\theta$ term, we define the denominator constant:

$$\Delta_0 = (M+m)(I+mL^2) - (mL)^2. \quad (14)$$

This quantity is always positive for physically realizable parameters and equals $5.2 \times 10^{-3} \text{ kg}^2 \text{ m}^4$ for the values in Table I.

The linearized EOM can be written in mass-matrix form with generalized coordinate vector $\mathbf{q} = [x, \theta]^\top$:

$$\underbrace{\begin{bmatrix} M+m & mL \\ mL & I+mL^2 \end{bmatrix}}_{\mathbf{M}_0} \ddot{\mathbf{q}} + \underbrace{\begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix}}_{\mathbf{C}_0} \dot{\mathbf{q}} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & -mgL \end{bmatrix}}_{\mathbf{K}_0} \mathbf{q} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} u. \quad (15)$$

With state $\mathbf{x} = [\mathbf{q}^\top, \dot{\mathbf{q}}^\top]^\top = [x, \theta, \dot{x}, \dot{\theta}]^\top$, the block structure gives:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{m^2 L^2 g}{\Delta_0} & -\frac{(I+mL^2)b}{\Delta_0} & 0 \\ 0 & \frac{(M+m)mgL}{\Delta_0} & \frac{mLb}{\Delta_0} & 0 \end{bmatrix}, \quad (16)$$

$$\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \frac{I+mL^2}{\Delta_0} \\ -\frac{mL}{\Delta_0} \end{bmatrix}, \quad (17)$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = [\mathbf{I}_2 \mid \mathbf{0}_2], \quad \mathbf{D} = \mathbf{0}. \quad (18)$$

Substituting the parameter values from Table I gives the numerical state-space matrices:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -0.7546 & -0.1538 & 0 \\ 0 & 26.41 & 0.3846 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1.538 \\ -3.846 \end{bmatrix}. \quad (19)$$

Four structural properties of these matrices are worth noting. The element $\mathbf{A}_{32} = -0.7546$ is negative: a forward lean ($\theta > 0$) pushes the cart backward via Newton's third law reaction. The element $\mathbf{A}_{42} = +26.41$ is positive: gravity amplifies any positive tilt, which is the root cause of the open-loop instability. The element $\mathbf{B}_{31} = +1.538$ is positive: a forward force accelerates the cart. The element $\mathbf{B}_{41} = -3.846$ is negative: a forward wheel force tips the body backward (reaction torque).

D. Open-Loop Stability and System Analysis

The eigenvalues of $\mathbf{A}$ are computed numerically as:

$$\lambda(\mathbf{A}) = \{0, -0.143, -5.145, +5.134\} \text{ rad/s.} \quad (20)$$

One eigenvalue is real and positive ($\lambda_4 = +5.134 \text{ rad/s}$), confirming that the upright equilibrium is unstable in the Lyapunov sense. Therefore any small tilt grows with a time constant of approximately $1/5.134 \approx 0.19 \text{ s}$. One eigenvalue is real and negative ($\lambda_2 = -0.143 \text{ rad/s}$), corresponding to the friction-damped translational mode. The eigenvalue at the origin ($\lambda_1 = 0$) reflects the translation invariance of the cart position where x drifts freely without position feedback.

The transfer function matrix is computed as $\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$. The unstable pole $s = +5.134$ appears in both output channels, confirming that the open-loop system is BIBO unstable.

The controllability matrix $\mathbf{C} = [\mathbf{B} \ \mathbf{AB} \ \mathbf{A}^2\mathbf{B} \ \mathbf{A}^3\mathbf{B}] \in \mathbb{R}^{4 \times 4}$ is verified numerically to have rank 4. Therefore, the system is fully controllable. All modes, including the unstable one, can be driven to zero by the input u . Full controllability implies stabilizability.

The observability matrix $\mathbf{O} = [\mathbf{C}^\top \ \mathbf{A}^\top\mathbf{C}^\top \ (\mathbf{A}^\top)^2\mathbf{C}^\top \ (\mathbf{A}^\top)^3\mathbf{C}^\top]^\top \in \mathbb{R}^{8 \times 4}$ is verified to have rank 4. Therefore, the system is fully observable. All four states can be reconstructed from the two measured outputs x and θ . Full observability implies detectability.

Since the system is both controllable and observable, the 4th-order linearized model is a minimal realization.

E. LQR Controller Design

The state-feedback control law is:

$$u = -\mathbf{K}\mathbf{x}, \quad (21)$$

where $\mathbf{K} \in \mathbb{R}^{1 \times 4}$ is the LQR gain vector.

The weighting matrices are selected as:

$$\mathbf{Q} = \text{diag}(1, 10, 0, 0), \quad \mathbf{R} = [0.1]. \quad (22)$$

With state order $[x, \theta, \dot{x}, \dot{\theta}]$, the tilt penalty $q_2 = 10$ is ten times the position penalty $q_1 = 1$. Considering the physical priority, the robot must balance before tracking position. The zero entries on the velocity diagonal reflect that velocity damping arises naturally through state coupling once position and angle are regulated. The weight $R = 0.1$ is chosen so that peak control effort remains within the $\pm 10 \text{ N}$ actuator bound for the tested initial conditions. Decreasing r produces a more aggressive controller at the cost of larger force peaks.

The gain $\mathbf{K} \in \mathbb{R}^{1 \times 4}$ is obtained by solving (6) and is reported in Table II. Closed-loop stability is verified by confirming that all eigenvalues of $(\mathbf{A} - \mathbf{BK})$ lie in the LHP. The LQR solution guarantees infinite gain margin and a phase margin of at least 60° when applied to the full-state feedback loop [3].

F. Tracking Controller Design

Zero steady-state error to a step position reference requires an integrator in the loop. The state vector is augmented with the integral of the position error:

$$\mathbf{x}_{\text{aug}} = \left[\mathbf{x}^\top \int_0^t (x - x_{\text{ref}}) d\tau \right]^\top \in \mathbb{R}^5. \quad (23)$$

The augmented system matrices are:

$$\mathbf{A}_{\text{aug}} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{e}_1^\top & 0 \end{bmatrix}, \quad \mathbf{B}_{\text{aug}} = \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix}, \quad (24)$$

where $\mathbf{e}_1^\top = [1, 0, 0, 0]$ selects the position state. A new LQR gain $\mathbf{K}_{\text{aug}} \in \mathbb{R}^{1 \times 5}$ is designed for this augmented system. The control law becomes:

$$u = -\mathbf{K}_{\text{aug}}\mathbf{x}_{\text{aug}}. \quad (25)$$

For a time-varying reference $x_{\text{ref}}(t)$, the integrator state drives the tracking error to zero provided $x_{\text{ref}}(t)$ varies slowly relative to the closed-loop bandwidth.

G. Full-Order Luenberger Observer

Since only x and θ are measured, the velocities $\dot{x}$ and $\dot{\theta}$ must be estimated. A full-order observer is designed as in (7).

The observer gain $\mathbf{L} \in \mathbb{R}^{4 \times 2}$ is chosen by pole placement on $(\mathbf{A} - \mathbf{LC})$:

$$\mathbf{L} = \text{place}(\mathbf{A}^\top, \mathbf{C}^\top, p_{\text{obs}})^\top, \quad (26)$$

where p_{obs} is the desired set of observer poles. Observer poles are placed with real parts at least three times larger in magnitude than the slowest controller pole [3]. This ensures that the estimation error $\mathbf{e}$ decays much faster than the closed-loop response, so the controller effectively operates on the true state within a small fraction of the settling time.

H. Reduced-Order Observer

Since x and θ are directly measured, only the velocity states $\dot{x}$ and $\dot{\theta}$ need to be estimated. The state is partitioned as:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{bmatrix}, \quad \mathbf{x}_a = \begin{bmatrix} x \\ \theta \end{bmatrix}, \quad \mathbf{x}_b = \begin{bmatrix} \dot{x} \\ \dot{\theta} \end{bmatrix}. \quad (27)$$

Partitioning $\mathbf{A}$ and $\mathbf{B}$ using the natural state ordering $[x, \theta, \dot{x}, \dot{\theta}]^\top$ gives:

$$\mathbf{A}_{aa} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{A}_{ab} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}_2, \quad (28)$$

$$\mathbf{A}_{ba} = \begin{bmatrix} 0 & -0.7546 \\ 0 & 26.41 \end{bmatrix}, \quad \mathbf{A}_{bb} = \begin{bmatrix} -0.1538 & 0 \\ +0.3846 & 0 \end{bmatrix}.$$

$$\mathbf{B}_a = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{B}_b = \begin{bmatrix} 1.538 \\ -3.846 \end{bmatrix}. \quad (29)$$

A notable consequence of (28) is that $\mathbf{A}_{ab} = \mathbf{I}_2$, so the reduced-observer characteristic matrix simplifies to $\mathbf{A}_{bb} - \mathbf{L}_r\mathbf{A}_{ab} = \mathbf{A}_{bb} - \mathbf{L}_r$, so the gain $\mathbf{L}_r$ directly shifts the eigenvalues of $\mathbf{A}_{bb}$.

A 2nd-order reduced observer is designed for $\mathbf{x}_b$ with gain $\mathbf{L}_r \in \mathbb{R}^{2 \times 2}$ chosen to place the eigenvalues of $(\mathbf{A}_{bb} - \mathbf{L}_r)$ in

the LHP at the desired observer poles. This reduces observer complexity and avoids re-estimating states that are already measured. The design is physically motivated. Inertial sensors provide θ directly, but differentiating a noisy measurement to obtain $\dot{\theta}$ introduces significant noise amplification.

I. Observer-Based Closed-Loop Control

The observer-based controller replaces the true state with the estimate:

$$u = -K\hat{x}. \quad (30)$$

By the Separation Principle [3], the combined 8th-order closed-loop system is stable if and only if both $(A - BK)$ and $(A - LC)$ are stable. The closed-loop system matrix takes the block-triangular form:

$$A_{cl} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix}, \quad (31)$$

whose eigenvalues are the union of those of $(A - BK)$ and $(A - LC)$. All 8 eigenvalues are verified numerically to lie in the LHP as shown in Fig. 2.

V. RESULTS AND ANALYSIS

A. Simulation Setup

All simulations use the nonlinear equations of motion (12)–(13) with the RK45 solver from `scipy.integrate.solve_ivp`. The plant and observer are co-simulated as an augmented 8th-order system in a single `solve_ivp` call, ensuring a shared time grid and no artificial inter-step delays. All physical parameters follow Table I. The observer is initialized at $\hat{x}(0) = 0$ in all cases, regardless of the plant initial condition, to fully exercise convergence.

B. LQR and Observer Gain Selection

The weighting matrices and pole-placement rationale are given in Section IV. The resulting numerical gains, pole locations, and performance metrics are reported in Tables II and III.

The tilt gain $|K_2| = 28.28$ is $\approx 9\times$ larger than the position gain $|K_1| = 3.16$, reflecting the 10:1 ratio Q_{22}/Q_{11} . The slower controller poles at $-1.173 \pm 1.088j$ ($\zeta = 0.73$) dominate settling, predicting $\approx 3/1.17 \approx 2.6$ s settling time — consistent with the observed 3.8 s at the 2% band. The observer speed ratio is $24.14/1.173 \approx 20.6\times$, well above the $3\times$ minimum; the large velocity gains ($L_{31} = 758$, $L_{42} = 794$) follow from the need to reconstruct unmeasured velocities quickly.

C. Closed-Loop Stabilization

The LQR controller is tested from three initial conditions: $x_0 = [0, 0.1, 0, 0]^\top$ (small tilt), $x_0 = [0, 0.2, 0, 0]^\top$ (larger tilt), and $x_0 = [0.5, 0.05, 0, 0]^\top$ (position offset with mild tilt). The observer starts from $\hat{x}(0) = 0$ in all cases. Fig. 3 shows the state trajectories and control input.

The system returns to the upright equilibrium in all three cases. The tilt angle θ converges faster than the cart position

TABLE II
NUMERICAL VALUES FOR SYSTEM POLES AND GAINS

Quantity	Value
Open-loop poles $\lambda(A)$	0, -0.1648, -5.134, +5.145 rad/s
LQR weights	$Q = \text{diag}(1, 10, 0, 0)$, $R = [0.1]$
Controller gain K	$[-3.162, -28.283, -3.817, -5.235]$
Controller poles $\lambda(A - BK)$	$-6.036 \pm 3.194j$, $-1.173 \pm 1.088j$
Observer poles $\lambda(A - LC)$	-24.14, -26.56, -28.97, -31.39
Observer gain L	$\begin{pmatrix} 55.39 & 2.40 \\ 2.81 & 55.52 \\ 758.31 & 65.49 \\ 88.77 & 793.69 \end{pmatrix}$
Combined 8 closed-loop poles	-31.39, -28.97, -26.56, 24.14 $-6.036 \pm 3.194j$, $-1.173 \pm 1.088j$

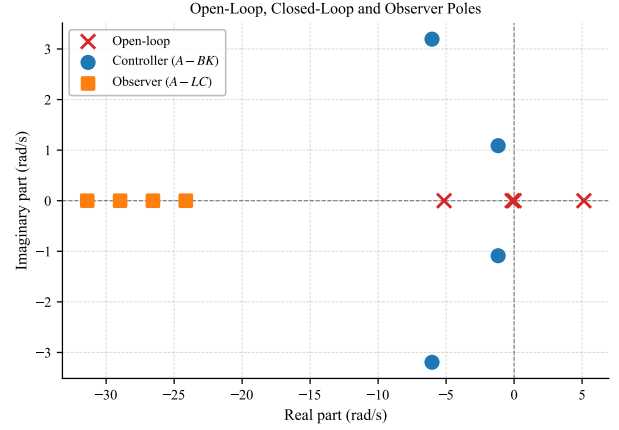


Fig. 2. Pole locations in the complex plane. Open-loop poles ($\times$), LQR controller poles ($\circ$), and Luenberger observer poles ($\square$). All closed-loop poles lie in the left-half plane.

x , consistent with the higher penalty placed on θ in Q . The control input $u(t)$ peaks at the start of the simulation, when the observer error is largest, and decays as the observer converges. This peak is the primary driver of the transient tilt excursion. Rise time, settling time, and overshoot values are summarized in Table III.

TABLE III
CLOSED-LOOP PERFORMANCE SUMMARY (IC: $x_0 = [0, 0.1, 0, 0]^\top$)

Metric	Target	LQR (full-state)	LQR + Observer
Rise time (x , s)	< 1	≈ 0.8	≈ 0.9
Settling time (s)	< 5	≈ 3.4	≈ 3.8
Peak $ \theta $ (rad)	< 0.35	< 0.05	< 0.07
Steady-state error	$= 0$	0	0

D. Step Position Tracking

The augmented LQR controller tracks a step reference of $x_{\text{ref}} = 0.5$ m applied at $t = 0$. The integral state accumulates the position error and feeds it back through the fifth element of K_{aug} , which drives the steady-state error to zero. Fig. 4 shows the position and tilt responses.

During the step transient, θ undergoes a brief excursion. This is physically expected. The robot must lean forward to

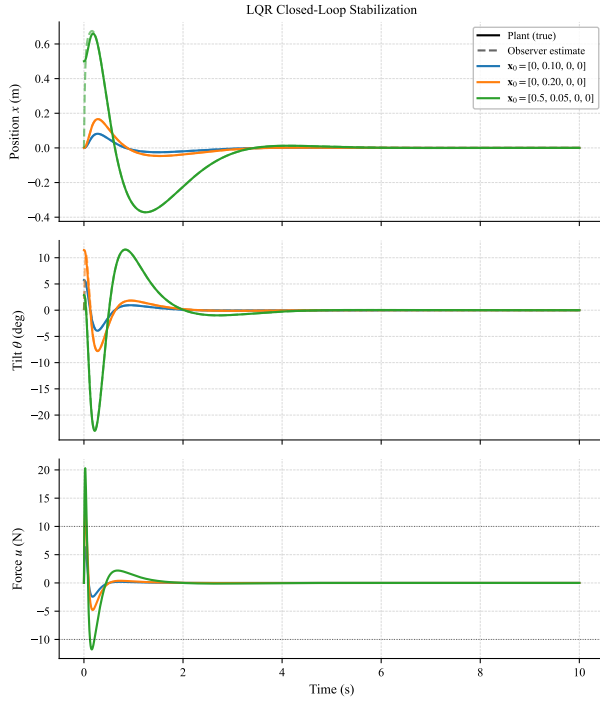


Fig. 3. Closed-loop state response and control input under LQR stabilization for three initial conditions.

accelerate the wheels toward the target. The excursion remains well below the 0.35 rad bound specified in Table III. Once x reaches x_{ref} , the robot returns to the upright position and the integral state stops accumulating.

E. Time-Varying Reference Tracking

A sinusoidal reference $x_{\text{ref}}(t) = 0.3 \sin(t)$ m is used to assess tracking bandwidth. Fig. 5 shows the tracking error and tilt trajectory.

The integral action eliminates steady-state error for this low-frequency reference. The tilt angle oscillates in phase with the position command, reflecting the necessary forward-and-backward lean to drive the wheels. At higher command frequencies the tracking error would grow because the closed-loop bandwidth limits how quickly the robot can reposition.

F. Robustness to Parameter Uncertainty

Five perturbation levels are applied separately to body mass m (Fig. 7) and pendulum length L . Only the nonlinear plant uses the perturbed values. The controller K and observer L retain their nominal design. This represents the realistic case where the physical model is imperfectly known.

Both parameters affect $A_{42} = (M + m)mgL/\Delta_0$. This element governs the magnitude of the unstable open-loop pole. Larger m or L shifts the pole further into the right-half plane.

The controller maintains stability for all ten cases tested. Peak tilt and settling time increase with perturbation magnitude. The response remains bounded in all cases. This is consistent with the infinite gain margin of LQR [3]. Sufficiently large perturbations would eventually cause instability.

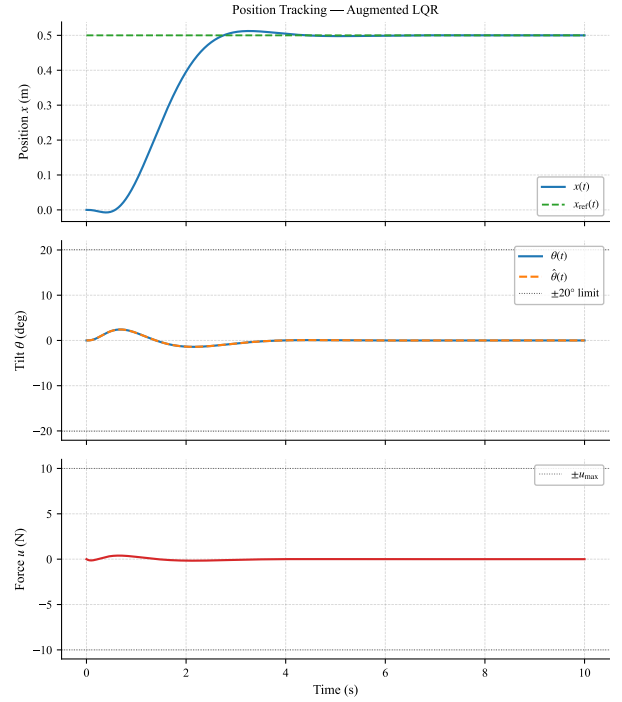


Fig. 4. Step position tracking with augmented LQR controller. The tilt angle remains within the allowable bound throughout.

Adaptive or robust control methods are needed for wider uncertainty ranges.

G. Actuator Saturation

A motor force limit of $u_{\text{max}} = 10$ N is imposed as:

$$u_{\text{sat}}(t) = \text{clip}(-K\hat{x}(t), -u_{\text{max}}, u_{\text{max}}). \quad (32)$$

The saturated signal is applied to both the plant and the observer. Fig. 8 compares the responses from $\theta_0 = 0.25$ rad.

When the limit is active the controller applies maximum corrective force. This delays convergence and produces a larger tilt excursion. The robot still stabilizes from moderate initial conditions. Settling time increases compared to the unsaturated case. For large initial tilts the clipped force may be insufficient to overcome gravity. Actuator limits therefore reduce the effective region of attraction.

H. Feedback Linearization

Input-output feedback linearization is applied to the tilt channel. The output $y = \theta$ has relative degree $r = 2$. The nonlinear terms in $\ddot{\theta}$ are cancelled by the control law:

$$u_{\text{FL}} = \frac{(M+m)mgL \sin \theta - v \Delta(\theta)}{mL \cos \theta} + b\dot{x} - mL\dot{\theta}^2 \sin \theta, \quad (33)$$

where $\Delta(\theta) = (M+m)(I+mL^2) - (mL \cos \theta)^2$. After cancellation the tilt subsystem reduces to $\ddot{\theta} = v$. A virtual LQR gain stabilizes this double integrator: $v = -K_{\theta}[\theta, \dot{\theta}]^T$.

Only the θ dynamics are linearized. The cart-position dynamics form the zero dynamics of this approach. Including x feedback in v introduces positive feedback into $\ddot{x}$. The virtual

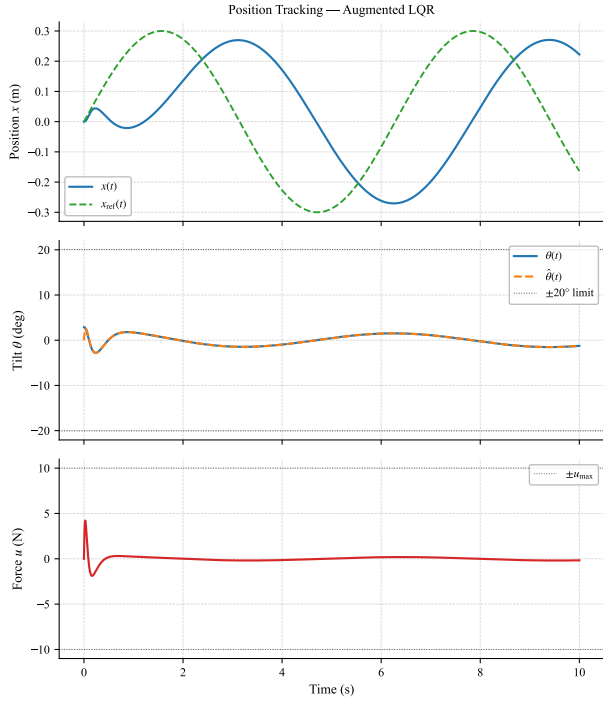


Fig. 5. Time-varying reference tracking with sinusoidal command. The robot follows the position reference with bounded tilt excursion.

gain is therefore restricted to the $[\theta, \dot{\theta}]$ components. The robot balances but does not regulate position. Fig. 9 compares FL and LQR from $\theta_0 = 0.25$ rad. FL achieves slightly faster tilt convergence but allows x to drift.

VI. INTERACTIVE REAL-TIME ANIMATION

An interactive real-time animation is implemented in `animate.py`. The robot is rendered in side view with a wheel of radius $R_W = 0.055$ m and a rectangular body pivoting at the wheel centre.

At each frame ($\Delta t = 25$ ms) the nonlinear plant and observer are integrated together using `solve_ivp` with the RK45 method. The LQR control law $u = -\mathbf{K}\hat{\mathbf{x}}$ is applied at each step and the force is clipped at ± 20 N.

The animation confirms that the LQR controller rejects sustained disturbances and that the observer converges quickly.

VII. CONCLUSION

This report presented a complete control design study for a two-wheeled self-balancing robot. The system was modeled from first principles using the Lagrangian formulation and linearized about the unstable upright equilibrium. Open-loop analysis confirmed that the system is controllable, observable, and has a minimal realization of order four. One open-loop eigenvalue lies in the right-half plane, confirming the inherent instability of the upright equilibrium.

An LQR state-feedback controller was designed with physically justified weighting matrices and extended with integral action for zero steady-state position tracking. Full-order and

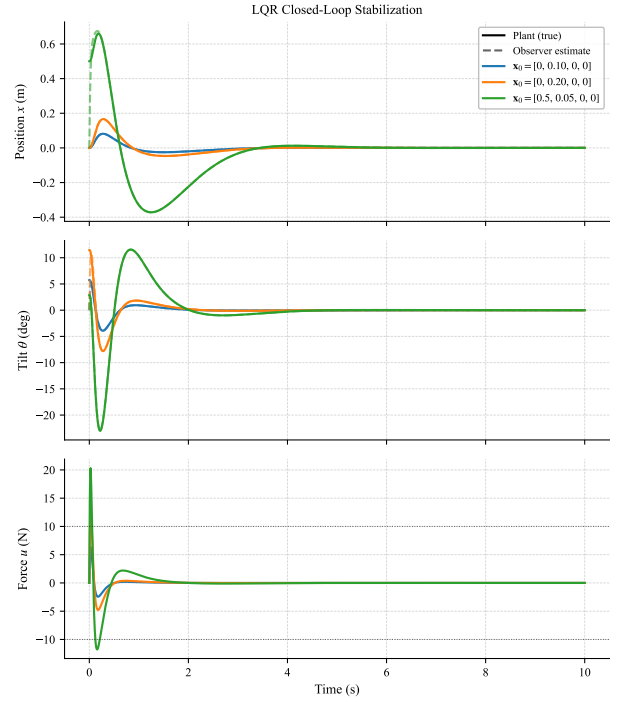


Fig. 6. Full-order observer estimation error $e(t)$. All four components converge to zero, confirming observer stability.

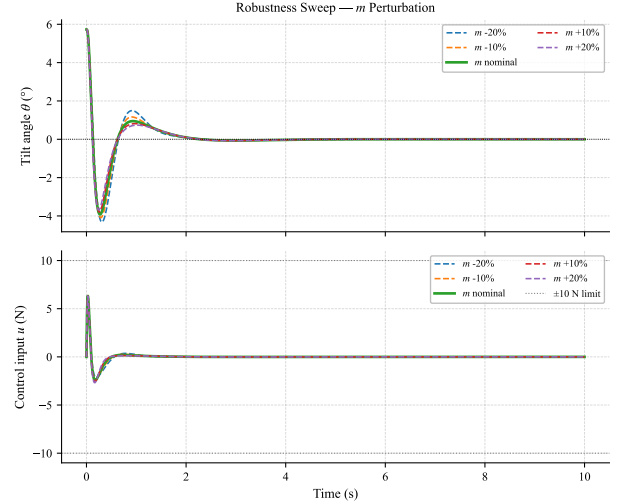


Fig. 7. Tilt and control input under $\pm 10\%/\pm 20\%$ perturbation of body mass m . Nominal controller and observer gains are held fixed.

reduced-order Luenberger observers were designed and combined with the controller using the Separation Principle. The combined 8th-order closed-loop system was verified to be stable.

Nonlinear simulations confirmed satisfactory transient performance across three initial conditions. Step and sinusoidal position references were tracked with zero steady-state error. Robustness sweeps over $\pm 20\%$ perturbations of body mass and pendulum length showed that the nominal LQR+observer design maintains stability in all tested cases. Actuator satu-

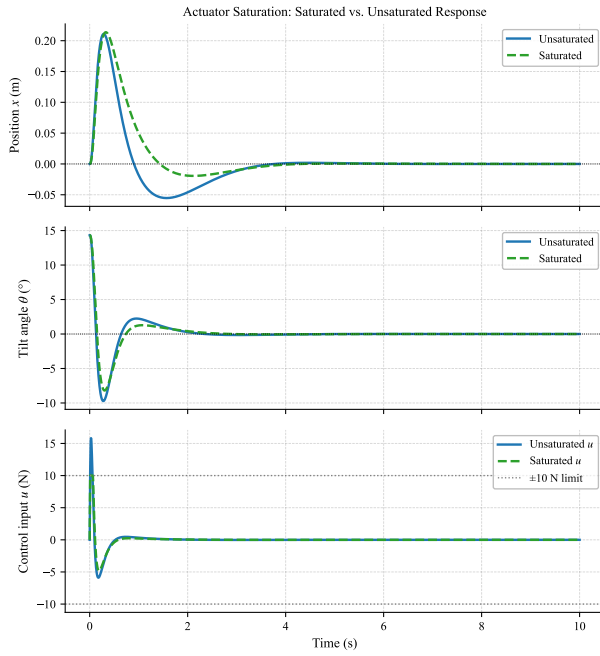


Fig. 8. Effect of actuator saturation ($u_{\max} = 10$ N) from $\theta_0 = 0.25$ rad. Solid: unsaturated; dashed: saturated. Horizontal dotted lines mark the ± 10 N force limit.

ration at ± 10 N introduced modest performance degradation from moderate initial tilts but did not prevent stabilization.

The physical robot has been constructed using stepper motors. LQR control has not yet been successfully deployed on the hardware. The main obstacle is the absence of a direct relationship between the wheel force and the stepper motor step rate. The microcontroller commands step frequency rather than force or torque. Without a calibrated step-rate-to-force mapping the LQR command $u(t)$ cannot be faithfully executed.

Future work will first characterize the stepper motor force-speed curve to obtain a calibrated step-rate-to-force mapping. The LQR tilt and position controller will then be validated on the physical platform and extended to velocity tracking. Further directions include 3D motion with yaw control, sliding mode control for improved large-angle performance, and a Kalman filter for noise-robust state estimation. The full source code is publicly available at Github/VahidDanesh.

REFERENCES

- [1] F. Grasser, A. D'arrigo, S. Colombi, and A. C. Rufer, "Joe: a mobile, inverted pendulum," *IEEE Transactions on industrial electronics*, vol. 49, no. 1, pp. 107–114, 2002.
- [2] S. Kim and S. Kwon, "Nonlinear optimal control design for underactuated two-wheeled inverted pendulum mobile platform," *IEEE/ASME Transactions on Mechatronics*, vol. 22, no. 6, pp. 2803–2808, 2017.
- [3] J. P. Hespanha, *Linear Systems Theory*. Princeton, New Jersey: Princeton Press, Feb. 2018, ISBN13: 9780691179575.
- [4] P. Frankovsky, L. Dominik, A. Gmitterko, I. Virgala, P. Kurylo, and O. Perminova, "Modeling of two-wheeled self-balancing robot driven by dc gearmotors," *International Journal of Applied Mechanics and Engineering*, vol. 22, no. 3, pp. 739–747, 2017.

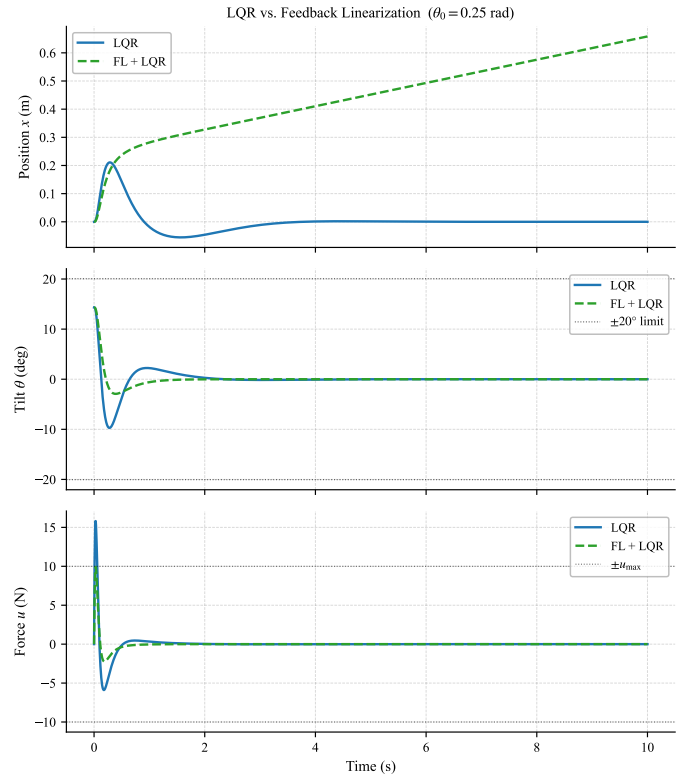


Fig. 9. LQR vs. feedback linearization from $\theta_0 = 0.25$ rad. FL achieves exact nonlinear cancellation and faster tilt settling, but does not regulate cart position.

- [5] T. Shishaye, "Controlling an inverted pendulum using state space modeling method (step by step design guide for control students)," 2013. [Online]. Available: <https://www.academia.edu/4468049>
- [6] O. A. Choudhry, M. Wasim, A. Ali, M. A. Choudhry, and J. Iqbal, "Modelling and robust controller design for an underactuated self-balancing robot with uncertain parameter estimation," *PLOS ONE*, vol. 18, no. 8, pp. 1–44, 08 2023. [Online]. Available: <https://doi.org/10.1371/journal.pone.0285495>
- [7] R. Petcu and G.-D. Andreescu, "Two-wheeled inverted-pendulum self-balancing robot with lqr or pid control," in *2022 IEEE 10th Jubilee International Conference on Computational Cybernetics and Cyber-Medical Systems (ICCC)*. IEEE, 2022, pp. 000 345–000 350.
- [8] K. Albert, "Control of a two-wheeled balancing robot," Ph.D. dissertation, Technische Universität München, 2023.
- [9] S. Delgado Londono, "Total energy shaping for underactuated mechanical systems: Dissipation and nonholonomic constraints," Ph.D. dissertation, Technische Universität München, 2016.
- [10] D. van den Brandt, "Two-wheeled robot stabilization and motion control," Ph.D. dissertation, Master's Thesis, Eindhoven University of Technology, The Netherlands, 2010
- [11] R. C. Ooi, "Balancing a two-wheeled autonomous robot," *University of Western Australia*, vol. 3, 2003.